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(Bisection - Method) OR (Bolzano Method)

EASY STUDY EASY SOLUTION

⇒ 1st step:

Interval: are given. If interval are not given than we assumed itself "2" intervals which give one value postive & one negative. For Frample

$x_0 = 0$ and $x_1 = 1$

⇒ 2nd step:

Put the intervals in given "f(x)" function.

Then put the value of x_0 & x_1 which give values one positive one negative

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then

here $f(a) \cdot f(b) < 0$ So the roots lies b/w 0 & 1. ——— (A)

EASY STUDY EASY SOLUTION

⇒ 3rd step:

To Find x_2 use formula:

$$x_2 = \frac{x_0 + x_1}{2} \quad \text{——— (i)}$$

Then put the value of x_2 in $f(x)$ function.

After putting value if function $f(x)$ give positive value then choose the above roots which already have negative value.

Otherwise

If $f(x)$ give negative value then choose the above roots which already have positive value.

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Then concluded the result as we conclued in eq (i).

EASY STUDY EASY SOLUTION

⇒ **4th step:**

To Find x_2 use formula:

$$x_2 = \frac{x_0 + x_1}{2} \quad \text{(ii)}$$

Then Pick the value of x_2 from (i) put in (ii).

After this put the value of x_2 in f(x) function.

Pin Ali Mehmood

Now steps are continuous which we disscuss in step 3.

MTH1005

(Bisection Method.)

Ex; Solve $x^3 - 9x + 1$ for the root between $x = 2$ and $x = 4$ by bisection Method.

Solution;

interval $(2, 4)$

$f(x) = x^3 - 9x + 1 = 0$

$f(2) = (2)^3 - 9(2) + 1 \Rightarrow 8 - 18 + 1 = -9$

$f(4) = (4)^3 - 9(2) + 1 \Rightarrow 64 - 36 + 1 = 29$

Here $f(2) f(4) < 0$ so Root lies between 2 and 4.

$x_0 = 2, \quad x_1 = 4$

$x_2 = \frac{x_0 + x_1}{2} = \frac{2 + 4}{2} = 3$

$f(3) = (3)^3 - 9(3) + 1 = 29 - 27 + 1 = 1$

Here $f(2) f(3) < 0$ so the Roots lies between 2 and 3.

$x_3 = \frac{x_0 + x_2}{2} = \frac{2 + 3}{2} = 2.5$

$f(2.5) = (2.5)^3 - 9(2.5) + 1 = -5.875 < 0$

So the Root lies between 2.5 & 3 as $f(2.5) f(3) < 0$

$x_4 = \frac{2.5 + 3}{2} = 2.75$ ——

EASY STUDY EASY SOLUTION
Pir Ad Muhammad

Ex#02; Carry out Five iterations for the function $f(x) = 2x\cos(2x) - (x+1)^2$

let interval $(-1, 0)$

$$f(x) = 2x\cos(2x) - (x+1)^2$$

$$f(-1) = 2(-1)\cos(-2) + (-1+1)^2$$

$$= -2(-0.4161) = 0.8322 > 0$$

$$f(0) = 2(0)\cos(0) - (0+1)^2 = -1 = -1 < 0$$

So the roots lie between 0 and (-1) as

$$f(0)f(-1) < 0$$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{0-1}{2} = -0.5$$

$$f(-0.5) = 2(-0.5)\cos(-1) - (-0.5+1)^2 = -0.790300$$

Roots lie between $f(-1)f(-0.5) < 0$

$$x_3 = \frac{x_0 + x_2}{2} = \frac{-0.5-1}{2} = -0.75$$

Same Find x_4, & x_5

Example; Carry out first five iterations

$$f(x) = x\cos x - 2x^2 + 3x - 1$$

$$1.2 \le x \le 1.3$$

$$f(1.2) = (1.2)\cos(1.2) - 2(1.2)^2 + 3(1.2) - 1$$

$$= 0.154829$$

$$f(1.3) = (1.3)(\cos)(1.3) - 2(1.3)^2 + 3(1.3) - 1$$

$$= -0.13225$$

So Roots lie between 1.2 & 1.3

$$f(1.2)(f)(1.3) < 0.$$

$$x_0 = 1.2, \quad x_1 = 1.3$$

$$x_2 = \frac{x_0 + x_1}{2} \Rightarrow \frac{1.2 + 1.3}{2} = 1.25$$

EASY STUDY EASY SOLUTION
Pir And Muhammad

(2) $f(1.25) = (1.25)\cos(1.25) - 2(1.25)^2 + 3(1.25) - 1$

$= = 0.01917 > 0$

$f(1.25) f(1.3) < 0$ so the Roots lie between 1.25 and 1.3

$x_3 = \frac{1.25 + 1.3}{2} = \frac{1.275}{2}$ ~~0.025~~

$f(-0.025) = (-0.025)\cos(-0.025) - 2(-0.025)^2 + 3(-0.025) - 1$

$= -0.0545 < 0$

$f(1.25)$ & $f(1.275) < 0$ so the Root lie between both

$x_4 = \frac{1.25 + 1.275}{2} = 1.2625 > 0$

$f(1.2625) =$ put in function

$= -0.0545 < 0$

$f(1.25) f(1.275) < 0$ so Root lie between both.

$x_5 = \frac{1.25 + 1.2625}{2} = 1.25625$

$f(1.25625) =$ ——— $= 0.00108 > 0$

$f(1.25625) f(1.2625) < 0$ so Roots lie between both;

$x_6 = \frac{1.25625 + 1.2625}{2} = 1.259375.$

EASY STUDY EASY SOLUTION
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Regula-Falsi Method
OR
Method of False Position

EASY STUDY EASY SOLUTION

Step 1:

Same as biseiton method.

Step 2:

Here put the value of x_1 & x_2 in formula:

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

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When we put the interval x_1 & x_2 in $f(x)$ function we find $f(x_1)$ & $f(x_2)$.

Next procedure is same as Biseiton method.

★ Regula-Falsi Method (Method of false position)

Formula;

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

EASY STUDY EASY SOLUTION

Ex:- use Regula Falsi Method to compute a Real Root of the equation $x^3 - 9x + 1 = 0$ if Roots lie between 2 & 4.

Sol; $x_1 = 2, \quad x_2 = 4$

$f(x_1)$: $f(2) = 2^3 - 9(2) + 1 = -9$

$f(x_2)$: $f(4) = 4^3 - 9(4) + 1 = 29$

For $n = 2$

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

$$x_{2+1} = x_2 - \frac{x_2 - x_{2-1}}{f(x_2) - f(x_2 - 1)} f(x_2)$$

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 4 - \frac{4 - 2}{29 - (-9)} (29) = 4 - \frac{2(29)}{38} = 2.4736$$

Pin And Muhammad

$$f(x_3) = f(2.4736) = 2.4736^3 - 9(2.4736) + 1 \Rightarrow$$
$$= -6.12644$$

$f(x_2) f(x_3) < 0$ Roots lie between x_2 & x_3

(3) Put $n = 3$

$$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3)$$

$$= 2.4736 - \frac{2.4736 - 4}{-6.12644 - 29} (-6.12644)$$

$$x_4 = 2.73989$$

$$f(x_4) = f(2.73989) = 2.73989^3 - 9(2.73989) + 1 = -3.090707$$

Since $f(x_4) f(x_2) < 0$ so Roots lie between x_4 & x_2.

Next apporimation is obtaind from, put $n = 4$

$$x_5 = x_4 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} f(x_4)$$

$$= 2.73989 - \frac{2.73989 - 2.4736}{-3.090707 + 6.12644} (-3.090707)$$

$$x_5 = 3.01100$$

$$f(x_5) = (3.01100)^3 - 9(3.01100) + 1 = 1.19909$$

EASY STUDY EASY SOLUTION
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Difference Between Bisection & Regula Falsi

In Bisection we have to choose Interval when $f(x_0)f(x_1) < 0$. And Important difference is that in Bisection we use last Approximation value for Finding the Interval of Next approximation.

But in Regula Falsi we have to use every Time 1st $f(x)$ in order to Find the Next approximation.

And Formula for Finding Iterations in Regula Falsi Method is

$$f(x_{n+1}) = x_n - \frac{x_n - x_{n-1}}{f(x_n) - F(x_{n-1})} f(x_n)$$

* In Regula Falsi when cos, sin and e functions includes in Given $f(x)$ we have to take x_0 and x_1 as Starting Interval.

and when ploynomial and Quardatic equation is given we have to take; x_1 and x_2 as a Starting Interval.

EASY STUDY EASY SOLUTION
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Method of Iteration

Step 1:

First $f(x)=0$ is rewrite in the form

$$x = \phi(x)$$

Step 2:

let two intervals $f(0)$ & $f(1)$ Put in $x=\phi(x)$ which gives one value >0 and one <0.

So, Root lie blw 0 & 1.

Step 3:

Take the derivate of function $x=\phi(x)$ which become $\phi'(x)$.

Put the value of $f(0)$ & $f(1)$ in $\phi'(x)$ which give value >0 & <0.

EASY STUDY (EASY SOLUTION)

✱ Method of Iteration

Method of Iteration can be applied to Find a Real Root of the equation $f(x)=0$ by rewriting the system in the form.

$$x = \phi(x)$$

Example;

$$f(x) = \cos x - 2x + 3 = 0$$

It can be; Rewritten as; $x = \frac{1}{2}(\cos x + 3) = \phi(x)$

$$\phi(x) = \frac{1}{2}(\cos x + 3) \rightarrow (1)$$

$$f(x) = \cos x - 2x + 3 = 0$$

$$f(1) = \cos(1) - 2(1) + 3 = 0 = 1.54030 > 0$$

$$f(2) = \cos(2) - 2(2) + 3 = 0 = -0.04161 < 0$$

So Roots lies between 1 & 2

$$\phi'(x) = -\frac{1}{2}(\sin x)$$

$$\phi'(1) = -\frac{1}{2}\sin(1) = -0.4207$$

$$\phi'(2) = -\frac{1}{2}\sin(2) = -0.4207$$

$\phi'(1)$ & $\phi'(2) < 1$ so the Method of iteration can be applied;

let $x_0 = 1.5$

$$x_1 = \frac{1}{2}(\cos x_0 + 3) = \frac{1}{2}(\cos(1.5) + 3) = 1.9998$$

$$x_2 = \frac{1}{2}(\cos x_1 + 3) = \frac{1}{2}(\cos(1.9998) + 3) = 1.9996$$

$$x_3 = \frac{1}{2}(\cos x_2 + 3) = \frac{1}{2}(\cos(1.9996) + 3) = 1.99969$$

So, this is the Required Roots upto 5 decimal places.

EASY STUDY EASY SOLUTION
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✱ Newton- Raphson Method

FORMULA USED FOR NEWTON RAPHSON METHOD;

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{for } n = 0, 1, 2, 3 \text{ ---}$$

✱ Newton's Raphson Method is also referred to as the "Method of tangents".

EASY STUDY EASY SOLUTION

Example :-

Find the real Root of the equation $x^3 - x - 1 = 0$ using Newton's-Raphson Method, Correct to Four decimal places.

Solution :-

$$f(x) = x^3 - x - 1$$

$$f(1) = (1)^3 - (1) - 1 = -1 < 0$$

$$f(2) = (2)^3 - (2) - 1 = 5 > 0$$

So the Roots lie between 1 & 2

Here $f'(x) = 3x^2 - 1$

$$f'(1) = 3(1)^2 - 1 = 2$$

$$f'(2) = 3(2)^2 - 1 = 11$$

So; $x_0 = 2$

The Second approximation is Computed using Newton - Raphson Method as;

$$N = 0 \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 2 - \frac{5}{11} = 1.54545$$

$$f(1.54545) = (1.54545)^3 - (1.54545) - 1$$
$$= 1.14576$$

$$f'(x) = 3x^2 - 1 = 3(1.54545)^2 - 1 = 6.16487$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.54545 - \frac{1.14576}{6.16487}$$
$$= 1.35961$$

EASY STUDY EASY SOLUTION

$$f(1.35961) = (1.35961)^3 - (1.35961) - 1$$
$$= 1.14576$$

$$f'(1.35961) = 3(1.35961)^2 - 1 = 6.16487$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.35961 - \frac{1.14576}{6.16487}$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = \text{---}$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} = 1.3247$$

Example:-

Set up Newton's scheme of iteration for finding the square root of a positive Number N.

Solution:-

The Square root of N can be carried out as a root of the equation;

EASY STUDY EASY SOLUTION

$$x^2 - N = 0$$

Let; $f(x) = x^2 - N$

By Newton's Method, we have;

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$f(x) = x^2 - N$; $f'(x) = 2x$

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Therefore;

$$x_{n+1} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$$

⋆ Newton's Method.

* In Newton's Method we have to put Min value in $f(x)$ like $(0, 1, 2, -1, -2)$ etc, we need Answer from Negative to positive or positive to Negative.
* IF Intervals are give use them in $f(x)$.
* Then we have to take derivative of given Function, and put the same values in $f'(x)$.
* Now The ~~valu~~ Solution is greater in both $f(x)$ and ~~or~~ $f'(x)$ use that value as a First approximation ; $x_0 =$ _____.

And then Continue Finding Iterations;

like;

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

* Put values of x_1 & x_2 — x_n in both Functions $f(x)$ and $f'(x)$ for Finding Next Iteration.

EASY STUDY EASY SOLUTION
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(Secant Method)

1st step:

We let the two intervals and then put in f(x) function.

x_0 & x_1

2nd step:

We do not need to check the interval that are ">0" & "<0". And also not to check the roots lie b|w the successive approximation in secant method.

3rd step:

To Find the value of x_2 use formula:-

EASY STUDY (EASY SOLUTION)
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For n=1

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

For n=2, 3, ... use this formula to find the next all iterations.

EASY STUDY ASSOCIATION
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Secant Method

The Secant method is Modified Form of Newton-Raphson Method.

" we replace $f'(x_n)$ by the difference ratio"

★ General form of Secant Method;

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

EASY STUDY EASY SOLUTION

Example;

Do three Iterations of secant Method to Find the root of;

$$f(x) = x^3 - 3x + 1 = 0$$

Taking; $x_0 = 1, \quad x_1 = 0.5$

★(we can also take (0, 1, -1, 2) But we have to choose Min value)★

$f(x_0) = f(1) = (1)^3 - 3(1) + 1 = -1$

$f(x_1) = F(0.5) = (0.5)^3 - 3(0.5) + 1 =$

for n = 1

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

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$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{(1)(-0.375) - (0.5)(-1)}{-0.375 - (-1)} = 0.2$$

For $n=2$

$$f(x_2) = f(0.2) = (0.2)^3 - 3(0.2)+1 = 0.408$$

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

$$= \frac{(0.5)(0.408) - (0.2)(-0.375)}{0.408 - (-0.375)} = 0.3563$$

For $n=3$

$$f(x_3) = f(0.3563) = (0.3563)^3 - 3(0.3563)+1$$

$$= -0.02367$$

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)}$$

$$= \frac{(0.2)(0.3563) - 0.3563(0.2)}{(0.3563) - (0.2)}$$

$$x_4 = 0.3473$$

Same Fiud:-

$$x_5 = -0.0000096$$

Anel At the enel we have to find dipperene absoulte value of Lest two Iterations;

Like;

$$|x_5 - x_4| = 0.0004$$

EASY STUDY EASY SOLUTION
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Muller Method

1st step:

It require three steps.

x_0, x_1 and x_2.

It does not require the derivative of function.

2nd step:

2nd we put the value of x_0, x_1 and x_2 in function given.

3rd step:

In Muller method we use formula:-

EASY STUDY EASY SOLUTION
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Muller's Method Formula:

$$x = x_0 - \frac{2c}{\pm b - \sqrt{b^2 - 4ac}}$$

Here we need the value of a, b & c which we obtained by using formula:-

$$c = f_0$$

$$b = \frac{f_1 - f_0 - ah_1^2}{h_1}$$

$$a = \frac{h_1 f_1 - (h_1 + h_2) f_0 + h_2 f_2}{h_1 h_2 (h_1 + h_2)}$$

Here we also need to find the value of h_1 & h_2 which we obtained from

$$h_1 = x_1 - x_0$$

$$h_2 = x_0 - x_2$$

By using these formula's we find 1st iteration. For 2nd iteration

EASY STUDY EASY SOLUTION
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4rth step:

For 2nd iteration

جو Muller's method نے فارمولا سے ہم

x_0 کی value کی $f(x)$

value وہ ہم کی (x_1) اور لے گے۔

لے گے جو question میں x_0 کی دی

گئی ہوگی۔ اور x_1 کی وہی لے گے

جو question میں دی گئی ہو۔

In this way we find 2nd iteration.

we also find the next iteration by using 2nd iteration concept.

EASY STUDY EASY SOLUTION
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✯ Muller's Method

=) Muller's Method is a generalization of the Secant Method.

=) It does not require the derivative of the Function. It is an Iterative Method that requires three starting points.

x_0, x_1, x_2.

EASY STUDY EASY SOLUTION

Example;

Do two Iterations of Muller's Method to solve $x^3 - 3x + 1 = 0$ Starting with;

$x_0 = 0.5$, $x_2 = 0$, $x_1 = 1$

Solution;

$$f(x_0) = f(0.5) = (0.5)^3 - 3(0.5) + 1 = -0.375$$

$$f(x_1) = f(1) = (1)^3 - 3(1) + 1 = -1$$

$$f(x_2) = f(0) = 0 - 3 \times 0 + 1 = 1$$

$$C = f(x_0) = -0.375$$

$$h_1 = x_1 - x_2 = 0.5$$

$$h_2 = x_0 - x_1 = 0.5$$

$$a = \frac{h_2 f_1 - (h_1 + h_2) f_0 + h_1 f_2}{h_1 h_2 (h_1 + h_2)}$$

$$= \frac{(0.5)(-1) - (-0.375) + (0.5)}{0.25}$$

$$a = 1.5$$

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$$b = \frac{f_1 - f_0 - a h_1^2}{h_1} = -2$$

∵ Formula for Finding Iterations in Muller's Method:

$$x = x_0 - \frac{2c}{+b - \sqrt{b^2 - 4ac}}$$

$$= 0.5 - \frac{2(-0.375)}{-2 - \sqrt{4 + 4(1.5)(0.375)}}$$

$$= 0.5 - \frac{-0.75}{-2 - \sqrt{4 + 2.25}} = 0.3333 < 0.5$$

For 2nd Iteration;

$x_2 = 0, \quad x_0 = 0.3333, \quad x_1 = 0.5$

$h_1 = x_1 - x_0 = 0.16667$

$h_2 = x_0 - x_2 = 0.3333$

$c = f_0 = 0.3333$

$f_0 = f(x_0) = 0.037046$

$f_1 = f(x_1) = -0.375$

$f_2 = f(x_2) = 1$

$$a = \frac{h_2 f_1 - (h_1 + h_2) f_0 + h_1 f_2}{h_1 h_2 (h_1 + h_2)} = \frac{0.023148}{0.027778}$$

$a = 0.8333$

EASY STUDY EASY SOLUTION
Pir Ali Muhammad

$$b = \frac{f_1 - f_0 - ah_1^2}{h_1} = -2.6$$

$$x = x_0 - \frac{2c}{b - \sqrt{b^2 - 4ac}}$$

$$= 0.333333 - \frac{0.0074092}{-5.2236}$$

$$= 0.3475 > 0.33333 = x_0$$

For 3rd ,

$x_2 = 0.33333$, $x_0 = 0.3475$, $x_1 = 0.5$

EASY STUDY EASY SOLUTION

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EASY STUDY EASY SOLUTION

Solution Of Linear

System of Equations

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Gaussian Elimination Method

EASY STUDY EASY SOLUTION

1st step:

Write the given system of equation in form

$$AX = B$$

Then

argumented matrix is

$$[A \mid B]$$

Step 2:

$$[A \mid B]$$

Making A reduce to upper triangular form

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Step 3:

Its reduce by using elementary transformation.

EASY STUDY EASY SOLUTION

Elementary transformation is applied according to given system of the equation.

Step 4:

After applied elementary transformation upper triangular system is solved by using back substitution process.

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Example

$$2x + 3y - z = 5$$
$$4x + 4y - 3z = 3$$
$$-2x + 3y - z = 1$$

EASY STUDY EASY SOLUTION

Sol:

step 1

$$\begin{bmatrix} 2 & 3 & -1 \\ 4 & 4 & -3 \\ -2 & 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}$$

step 2

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 4 & 4 & -3 & 3 \\ 2 & 3 & -1 & 1 \end{array}\right]$$

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step 3

Reduce to upper triangular form

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Divided the 1st equation by 2 and the substract equation (multiply by 4 and -2) from 2nd and 3rd equation. Thus, we eliminate x from 2nd and 3rd equation.

The new resulting equation is

$$\left.\begin{aligned} x + 3/2\,y - 1/2\,z &= 5/2 \\ -2y - z &= -7 \\ 6y - 2z &= 6 \end{aligned}\right\}$$

we divide 2nd equation by "-2" and eliminate y from 3rd equation. Thus new equation is

$$x + 3/2\,y - 1/2\,z = 5/2$$

$$y + 1/2\,z = 7$$

$$-5z = -15$$

EASY STUDY EASY SOLUTION
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With step

By Backward Substitution:

$$z = 3$$

Put value of "z" 2nd eq

$$y + \frac{3}{2} = \frac{1}{2} \Rightarrow y = \frac{7}{2} - \frac{3}{2}$$

$$y = \frac{4}{2} = 2$$

Put value of "y" and "z" in 1st eq

$$x + \frac{3}{2} \times 2 - 3/2 = 5/2$$

$$x + 3 - 3/2 = 5/2$$

$$x = \frac{5}{2} + \frac{3}{2} - 3 = \frac{5+3-6}{2}$$

$$x = 2/2$$

$$x = 1$$

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Gauss-Jordon Elimination Method

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Step 1:

The elements above and below the diagonal are made zero.

Step 2:

A equvilent diagonal form using elementaay transformation.

After this solution of resulting diagonal system obtained.

Example:

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$$x + 2y + z = 8$$

$$2x + 3y + 4z = 20$$

$$4x + 3y + 2z = 16$$

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Sol:

In matrix notation

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ 16 \end{bmatrix}$$

Now

$(-2)R_1 + R_2$ & $(-4)R_1 + R_3$ then

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & -5 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ -16 \end{bmatrix}$$

eliminat "y" from 1st & 3rd row using 2nd row

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & -1 & 2 \\ 0 & 0 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 4 \\ -36 \end{bmatrix}$$

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eliminat "z" from 1st and 2nd row using Normalized 3rd row

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

eliminat 'z' from above equation

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

Solution is

$x = 1$

$y = 2$

$z = 3$

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$$[L][U] = [A]$$

$$\begin{bmatrix} a & 0 & 0 \\ d & b & 0 \\ e & f & c \end{bmatrix} \begin{bmatrix} 1 & g & h \\ 0 & 1 & i \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & -1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a & ag & ah \\ d & dg+b & dh+bi \\ e & eg+F & eh+fi+c \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & -1 & 4 \end{bmatrix}$$

By compare

$\boxed{a=5}$, $ag = -2 \Rightarrow 5g = -2 \Rightarrow \boxed{g = -2/5}$

$ah = 1 \Rightarrow 5h = 1 \Rightarrow \boxed{h = 1/5}$, $\boxed{d = 7}$

$dg + b = 1 \Rightarrow$ By putting values we obtained $\boxed{b = \frac{19}{5}}$, $dh + bi = -5$ By putting values $\boxed{i = -\frac{32}{19}}$

$\boxed{e = 3}$

$eg + F = 3$ By putting value $\boxed{F = \frac{41}{5}}$

$eh + fi + C = 4$ By putting value:- $\boxed{C = \frac{327}{19}}$

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Put all value in [L] & [U]

$$[L][U][X] = [B]$$

$$\begin{bmatrix} 5 & 0 & 0 \\ 7 & 19/5 & 0 \\ 3 & 41/5 & 327/19 \end{bmatrix} \begin{bmatrix} 1 & -2/5 & 1/5 \\ 0 & 1 & -32/19 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

step 3 By Substituting

$$[L][Y] = B$$

$$\begin{bmatrix} 5 & 0 & 0 \\ 7 & 19/5 & 0 \\ 3 & 41/5 & 327/19 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

By backward substitution:

$5y_1 = 4 \Rightarrow \boxed{y_1 = \frac{4}{5}}$

$5y_1 + \frac{19}{5}y_2 = 8 \Rightarrow \boxed{y_2 = \frac{12}{19}}$

$3y_1 + \frac{41}{5}y_2 + \frac{327}{19}y_3 = 10 \Rightarrow \boxed{y_3 = \frac{46}{327}}$

step 4:

$$[U][X] = Y$$

$$\begin{bmatrix} 1 & -2/5 & 1/5 \\ 0 & 1 & -32/19 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 12/19 \\ 46/327 \end{bmatrix}$$

$x_1 - \frac{2}{5}x_2 + \frac{1}{5}x_3 = 4/5$

$x_1 = \left(\frac{2}{5}\right)\left(\frac{284}{327}\right) + \left(\frac{1}{5}\right)\left(\frac{46}{327}\right) = \frac{4}{5}$

$\boxed{x_1 = \frac{366}{327}}$

$x_2 - \frac{32}{19}x_3 = \frac{12}{19}$

$x_2 = \left(\frac{32}{19}\right)\left(\frac{46}{327}\right) = \frac{12}{19}$

$\boxed{x_2 = \frac{284}{327}}$

$\boxed{x_3 = \frac{46}{327}}$

$x_1 = \frac{366}{327}$, $x_2 = \frac{284}{327}$

$x_3 = \frac{46}{327}$

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Jacobi's Method of Matrix Inversion

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This method is good for

example.

$$8x - 3y + 2z = 20$$
$$4x + 11y - z = 33$$
$$6x + 3y + 12z = 35$$

Sol:=

First we arrange the system is diagonally dominant. and

let the Right side of equation

$x = x_0$

$y = y_0$

$z = z_0$

$$x = \frac{1}{8}[20 + 3y_0 - 2z_0]$$

$$y = \frac{1}{11}[33 - 4x_0 + z_0]$$

$$z = \frac{1}{12}[35 - 6x_0 - 3y_0]$$

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step 2:

For 1st iteration

we start with an initial approximation $x_0 = y_0 = z_0 = 0$

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Put in 1st step we obtained

$$x_1 = \frac{1}{8}[20 + 3(0) - 2(0)] = 2.5$$

$$y_1 = \frac{1}{11}[33 - 4(0) + 0] = 3$$

$$z_1 = \frac{1}{12}[35 - 6(0) - 3(0)] = 2.9167$$

step 3:

For 2nd iteration

$$x_2 = \frac{1}{11}[20 + 3y_1 - 2z_1]$$

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$$y_2 = \frac{1}{11}[33 - 4x_1 + z_1]$$

$$z_2 = \frac{1}{12}[35 - 6x_1 - 3y_1]$$

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Put value of x_1, y_1 and z_1 in Right side of equation we obtained the values of 2nd iteration.

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Similarly All the next iterations is find by understanding this concept of iteration.

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Gauss-Seidel Iteration Method

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$$10x + 2y - z - w = 3$$
$$-2x + 10y - z - w = 15$$
$$-x - y + 10z - 2w = 27$$
$$-x - y - 2z + 10w = -9$$

Sol:

$$x = \frac{1}{10}[3 + 2y + z + w]$$

$$y = \frac{1}{10}[15 + 2x + z + w]$$

$$z = \frac{1}{10}[27 + x + y + 2w]$$

$$w = \frac{1}{10}[-9 + x + y + 2z]$$

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step 2:

1st iteration

[Put (y = z = w = 0) on R.H.S of eq in step 1]

$x_1 = 0.3$

[Put (x = 0.3, z = w = 0) on R.H.S of eq in step 1]

$y_1 = 1.56$

[Put x = 0.3, y = 1.56 and w = 0 in R.H.S of eq in step 1]

$z_1 = 2.886$

[Put x = 0.3, y = 1.56 & z_1 = 2.886 in R.H.S of eq in step 1]

$w_1 = -0.1368$

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step 3:

For 2nd iteration

Pull all values of x_1, y_1, z_1, w_1 to find 2nd iterat

$$x_2 = \frac{1}{10}[3 + 2y_1 + z_1 + w_1]$$

$$y_2 = \frac{1}{10}[15 + 2x_1 + z_1 + w_1]$$

$$z_2 = \frac{1}{10}[27 + x_1 + y_1 + 2w_1]$$

$$w_2 = \frac{1}{10}[-9 + x_1 + y_1 + 2z_1]$$

step 4:

For next iterations follow the step 3.

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Eigen value Problems of Power Method

step 1:

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Choose the intial vector such that largest element is unity.

initial vector اس کا مطلب ہے کہ ہم سوال میں $[V^{(0)}]$ کو ظاہر کرتے ہیں اگر سوال میں $V^{(0)}(1,1,1)$ initial vector نہ دیا ہو تو ہم نے ہمیشہ لینا ہے $V^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ OR

یہ rule ہے Power method کا

step 2:

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$V^{(0)}$ is pre-multiplied by the matrix $[A]$

(ز) equation کو its means

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$V^{(0)}$ کا نام دے کر [A] سے گئی ہوگی ہم
pre-multiplied کریں گے۔
یعنی [A] کا مطلب ہے کہ pre-multiplied
(multiply) system A کو $V^{(0)}$ کو
کرنا ہے۔

$$U^{(1)} = [A]\, V^{(0)}$$

Step 3:

In Step 2,
$U^{(1)}$ (resultant vector) کو
ظاہر کرے گا۔

The resultant vector again normalized.

The result look at this Point is

$$U^{(k)} = [A]\, V^{(k-1)} = q_k V^{(k)}$$

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step 4:

In step 3

q_k denote desire largest value

$v^{(k)}$ Corresponding eigenvector.

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At the end, Process of iteration continued.

(λ) p_i کو value کی q_k سے replace کریں گے تمام

اور iteration کرنے کے بعد

(X) p_i کو value کی $v^{(k)}$ سے replace کریں گے

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Now, we do example of this method.

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Example

$$[A] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 9 \end{bmatrix}$$

1st step:

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Choose initial vector

$$v^{(0)} = (1, 1, 1)^T$$

Compute 1st iteration

$$u^{(1)} = [A][v^{(0)}] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \\ 14 \end{bmatrix}$$

Now, normalize the resultant vector

$$u^{(1)}$$

$$u^{(1)} = 14 \begin{bmatrix} 1/2 \\ 6/7 \\ 1 \end{bmatrix}$$

$$v^{(1)} = q_1 v^{(1)}$$

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step 2

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Relaxation Method

Step 1:

Equation کا system دیا گیا ہو گا۔ system of equation

کو L.H.S اور R.H.S پہ لے جائیں گے اور L.H.S والی

کو ہم denote سے R_x, R_y & R_z کریں گے۔

Step 2:

1st eq میں ہم دیکھیں گے کہ جو ("x") کی value ہے وہ

2nd eq کی (y, z) کی نسبت بڑی ہو۔ اسی طرح

3rd eq میں ("y") کی بڑی ہو (x, z) سے وغیرہ میں۔

میں ("z") میں کی بڑی ہو (x, y) سے وغیرہ میں۔

Otherwise

اگر equation میں ایسا نہ ہو تو ہم نے (1st eq) کو اس

سے eq replace کرنا ہے جس میں (x) کی value بڑی

ہو۔ اور دوسری میں (y) کی اور تیسری میں

(z) کی۔

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3rd step:

table 6 (Residuals increment) پھر ہم نے پہلے بنایا تھا

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Residuals increment	δR_x	δR_y	δR_z
δx			
δy			
δz			

1st step جو ہم نے value لکھنی ہے وہ اس میں لکھنی ہے جو ہم نے P.H.S میں x, y, z کی دی گئی ہو۔ 1st eq کی 1st line میں اسی طرح دوسری 2nd eq کی 2nd line اور 3rd eq کی 3rd line میں لکھنی ہے (Horizontal values) اور یہ vertical کی۔

step 4:

پھر بنائیں Relaxation Table اس

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2nd iteration

$$u^{(1)} = [A][v^{(1)}] = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 3 & 5 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1/2 \\ 6/7 \\ 1 \end{bmatrix} = \begin{bmatrix} 39/7 \\ 67/7 \\ 41/7 \end{bmatrix}$$

$$u^{(2)} = 12.2143 \begin{bmatrix} 0.45614 \\ 0.78362 \\ 12.2143 \end{bmatrix}$$

$$u^{(2)} = q_{(2)} v^{(2)}$$

step 3:

Continue this process to Follow the steps to find all next iteration.

Jacobies Method for finding
Eigen values and eigenvectors
I have added at the end of
This file

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Example:

$$9x - y + 2z = 9$$
$$x + 10y - 2z = 15$$
$$2x - 2y - 13z = -17$$

EASY STUDY EASY SOLUTION

$$9x - y + 2z = 9 \quad \text{(i)}$$
$$x + 10y - 2z = 15 \quad \text{(ii)}$$
$$2x - 2y - 13z = -17 \quad \text{(iii)}$$

$$R_x = 9 - 9x + y - 2z$$
$$R_y = 15 - x - 10y + 2z$$
$$R_z = -17 + 2x - 2y + 13z$$

Residual table or Operation table

number	R_x	R_y	R_z
δx	-9	-1	-2
δy	1	-10	+2
δz	-2	+2	13

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Relaxation table

	x	y	z	R_x	R_y	R_z
1.	0	0	0	9	13	-1
2	0	0	1.3	6.4	17.6	-0.1
3	0	1.76	1.3	8.16	0	3.42
4th	0.906	1.76	1.3	0.006	-0.906	1.608

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اس کا طریقہ یہ ہے کہ سب سے پہلے $x=y=z=0$ لیتا ہے جو R_x, R_y, R_z میں Put کرنا ہے۔ پھر جو values آئے گی اس میں سب سے بڑی value لینی ہے جیسا کہ اوپر Table میں R_z کی value سب سے بڑی ہیں۔ پھر ہم نے وہی value کو Operation table میں (z) میں دیکھنا ہے اور اگر وہ (+ive) ہے تو (+ive) لے گے اگر (-ive) ہے تو (+ive) لے گے۔ اور پھر بڑی (-ive) کو value "z" والی value سے تقسیم کرنا ہے۔ یعنی $\left(\frac{-1.6}{-10} = 1.3\right)$

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طریقہ (2nd approximation)

جبکہ $x=y=0$ ہوگا

R_x, R_y, R_z کی values معلوم کرنے کا

Procedure

والسے ہی Step کی طرح جاری رکھنا

ہے یہ Table کو اچھے سے سمجھ لیں کیسے

next all approx: کرنا ہے

اور یہ تب تک جاری رکھنا ہے جب تک

ہمیں (0) سے قریب ترین value نہ

مل جائیں R_x, R_y, R_z کی

THE END

Pin Study Muhammad

(Jacobi's Method)
(For Finding Eigen values)

Procedure;

① سب سے پہلے جو میٹرک دیا گیا ہوگا اس میٹرک کی سب سے بڑی off-diagonal ایلیمنٹ کو دیکھنا ہوتا ہے۔

② اگر off-diagonal میں بڑی ویلیو پہلی Row اور دوسرے کالم میں ہوگی تو اسکو ہم (a_{12}) کہو سکتے ہیں۔ ہمیشہ دو ویلیوز ایک جیسی ہوتی ہیں off-diag میں۔

③ اگر off-diagonal میں بڑی ویلیو دوسری Row اور دوسرے کالم میں ہوتو اسکو (a_{22}) کہو سکتے ہیں۔

Tips for using formula;

$$\tan 2\theta = \frac{2a_{ij}}{a_{ii} - a_{jj}}$$

اگر (a_{13}) ہوتو فرض کرلیں۔ $i=1, j=3$ ہر i اور j کی جگہ put کریں۔

$$\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$$

اسی طرح اگر a_{12} ہوتو، $i=1, j=2$

$$\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$$

اسی طرح اگر a_{23} ہوتو، $i=2, j=3$

$$\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$$

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(3) فارمولہ حل کرنے کے لیے θ کی ایک ویلیو آئے گی - اس کے لیے ایک orthogonal میٹرکس بنانا ہوتا ہے:-

(Trick for Making Matrix)

اگر فارمولا میں (a_{13}) استعمال کیا ہے تو اس کا مطلب ہے ایک اور تین - پہلی Row اور تیسرا کالم -

یا سکو ایسے سمجھ لیں پہلی اور تیسری Row کے Matrix میں diagonal میں $\cos\theta$ آئے گا - θ کو (a_{13}) کے حساب سے Matrix ایسے بنے گا:-

$$S_1 = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

(a_{12}) ہوگا تو اس کا مطلب پہلی Row اور دوسرا کالم

$$S_1 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a_{23}) ہوگا تو

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

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(4) کی ویلیوز orthogonal Matrix میں put کرنے کے بعد Rotations معلوم کرنا ہوتی ہیں ۔ مثال کے طور پر $D_1, D_2 \cdots$

ے اور رو ٹیشن تب تک معلوم کرنا ہوتی جب تک Matrix کی off-diagnals عام elements / entries زیرو نہ ہو جاتیں ۔

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Rotations معلوم کرنے کے لیے فارمولا :-

$$D_1 = S_1^{-1} A S_1$$

ے پہلی Rotation معلوم کرنے کے بعد جو Matrix آئے گی پھر اس Matrix کو استعمال کرتے اگلی Rotation معلوم کرنا ہوگی ۔

(5) ے جب آخری Rotation کے بعد جو Diagonal میں ویلیوز ہوں گی وہ اس Matrix کی Eigenvalues ہوں گی ۔

(6) ے Eigenvectors معلوم کرنے کے لیے جو Rotations نکالی ہوں گی ان Matrices کو آپس میں Multiply کریں گے تو Eigenvector آجائے گا ۔

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step 4:

Put the value of two interval f(0) & f(1) in

$$\phi'(x)$$

Then which value of $\phi'(x) < 1$, we let it

$$x_0 = x_0$$

(x0 کی وہ value لے گے جو ہم نے interval میں فرض کیے ہوگے)

Step 5: Then (ان interval کے درمیان والی رقم)

$$x_1 = \phi(x_0)$$

$$x_2 = \phi(x_1)$$

$$x_3 = \phi(x_2)$$

$$x_4 = \phi(x_4$$

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